Curriculum Vitae

Dr Wali Islam

PhD (UNSW), Graduate Diploma (UNSW), MBBS (Bangladesh)

Mucosal Immunology | Ethical Clinical Practice | Public Health

| Effective Communication | Community Development

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Professional Summary:

Dr Abul F. M. Wali Ul Islam is a distinguished medical doctor, biomedical scientist and academic with a multidisciplinary career spanning medicine, immunology, microbiology, occupational and environmental health, higher education and scientific research. He holds an MBBS from the University of Rajshahi, a Graduate Diploma in Microbiology and Immunology, and a PhD in Immunology from UNSW. His research has focused on infection and immunity, mucosal immunology, inflammatory disease and hepatobiliary–gastrointestinal interactions. He has held research and academic positions at UNSW and Prince of Wales Hospital, as well as international leadership roles including Global Vice-President of Sun Moon University in Korea and Visiting Academic and Research Fellow at Bangladesh University of Health Sciences.

Alongside his scientific career, Dr Wali Islam has extensive experience in international leadership, institutional development, strategic partnerships and community empowerment. Guided by strong ethical and moral principles, he promotes cultural competence and effective communication in healthcare, recognising the importance of patients' values, beliefs and social contexts. His work across Australia, Bangladesh and other regions has contributed to health and social development, education, occupational health and safety, humanitarian assistance and research capacity building, particularly to expand healthcare, education, scientific knowledge and vocational opportunities for disadvantaged and underserved communities.

A key strength of Dr Wali Islam's leadership is his ability to combine scientific expertise, ethical leadership, academic excellence and culturally responsive engagement. He is committed to building partnerships among universities, governments, professional institutions and international organisations to address complex global challenges through collaboration. With a particular focus on strengthening Australia–Bangladesh cooperation, he promotes knowledge exchange, innovation and institutional capacity for sustainable development. His overarching vision is to help build healthier, better educated, scientifically capable and socially empowered communities through excellence in healthcare, education, research and international partnership.

Education

PhD in Immunology – University of New South Wales (UNSW), Australia
WHO Scholar | Thesis awarded *High Distinction* (95%)

Graduate Diploma in Microbiology & Immunology – UNSW, Australia

MBBS – University of Rajshahi, Bangladesh
University Medallist

Citizenship

Bangladeshi (by birth) | Australian (Distinguished Talent Visa)

Professional Attributes

- Independent and collaborative researcher
- Strong academic writing and project development capability
- Effective communicator across scientific and community settings

Research Expertise

- Mucosal immunology and epithelial barrier function
- Hepatobiliary–gut immune crosstalk
- Inflammatory bowel disease (IBD) pathogenesis
- Host–microbiome interactions
- Enteric infection and bacterial translocation
- Cytokine biology (TNF-α, endotoxin-driven inflammation)
- Translational animal models of inflammation and infection

Major Scientific Contributions

Biliary Immune Regulation of the Intestinal Mucosa

First to demonstrate that inflammatory cytokines secreted into bile directly regulate gut mucosal integrity and immune responses.

Mechanistic Link Between Biliary Cytokines and Enteritis

- Provided causal in vivo evidence that biliary TNF-α induces enteritis, mucosal ulceration, haemorrhage, and epithelial disruption.

Bidirectional Gut–Liver Signalling in Systemic Disease

- Demonstrated that hepatobiliary immune activation amplifies systemic inflammation, contributing to sepsis and multiple organ dysfunction syndrome (MODS).
- Identified Kupffer cell–driven immune cascades as initiators of downstream mucosal pathology.

Luminal Control of Bacterial Translocation

- Showed that intraluminal conditions (microbial behaviour, bile composition, local environment) outweigh systemic immunity in determining bacterial translocation.
- Provided a conceptual framework relevant to dysbiosis, IBD, and infection-driven inflammation.

Validated Animal Models

- Acute infectious diarrhoea (live bacterial orogastric challenge)
- Endotoxin-induced systemic inflammation (IV/IP LPS)
- Whole bacterial antigen models to dissect mucosal vs systemic immunity

Translational Innovation

- Conceptualised **intraluminal neutralisation of biliary cytokines** (notably TNF-α) as a novel therapeutic strategy.

Future Research Program

1. **Next-Generation IBD Models**
    - Non-chemical, host–microbiome-driven models reflecting human disease complexity.
2. **Hepatobiliary Immune Modulators**
    - Roles of biliary cytokines and immunoglobulins in epithelial injury, dysbiosis, and inflammation.
3. **Microbiome–Immune Interactions**

- Luminal regulation of microbial virulence, growth, and barrier disruption.

4. **Gut–Brain and Gut–Liver Axes**
    - Neuroimmune regulation of motility, pain, and inflammation.
5. **Therapeutic Translation**
    - Intraluminal immune modulation, targeted probiotics, and mucosal repair strategies.

Academic Appointments

Current

- Visiting Academic & Research Fellow, Bangladesh University of Health Sciences (BUHS)
- Global Vice-President, Sun Moon University, South Korea

Previous

- Academic, School of Medical Sciences, UNSW (5+ years)
- Scientist, Prince of Wales Hospital, Sydney (3+ years)

Teaching, Supervision & Mentorship

- Supervision of Honours, Master's, and PhD-level research projects
- Founder, *Study Support Australia* – mentoring international and HDR students
- Strong commitment to capacity building and inclusive research training

Leadership and Service

- Extensive experience in interdisciplinary collaboration across immunology, microbiology, hepatology, and GI physiology
- Active contributor to global health research and Low- and Middle-Income Countries capacity building

In conclusion, Dr Abul F. M. Wali Ul Islam's career embodies a rare integration of medical and scientific expertise, academic leadership and international development. His contributions across healthcare, biomedical research, higher education and community advancement demonstrate a consistent ability to translate knowledge into meaningful action, strengthening systems and expanding opportunities for underserved populations.

Looking ahead, Dr Wali Islam is well positioned to deepen international cooperation—particularly between Australia and Bangladesh—by fostering partnerships that drive innovation, knowledge exchange and sustainable institutional growth. His commitment to ethical leadership, cultural understanding and collaborative problem-solving continues to guide his work.

Through this integrated approach, he aims to help build healthier, better educated, scientifically capable and socially empowered communities, advancing long-term progress locally, nationally and globally.

Please do not hesitate to contact for any further information,

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